Covariance-defect correction and transported mean innovations for white-box MLP estimation

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Phase 2 technical writeup · Submission **336809**
Algorithm ALG058-V001 · Review draft, 18 October 2026

Abstract

We describe the estimator submitted as **336809**, which attained a public budget-adjusted final-layer mean squared error of $2.97110640613 \times 10^{-9}$, with raw error $1.94467643766 \times 10^{-8}$. It builds on the published factored cumulant engine of 504aldo and LordProcciuto. Our additions are a fitted, analytically motivated covariance-defect dictionary inside the physical recurrence, and small learned local mean innovations propagated through a separate linear response recurrence. The final model averages two fixed innovation sources, preserving their separate feature normalizations in one batched evaluation.

On four exposed development networks, our complete adaptation reduces adjusted error by **4.55%** relative to the unmodified public engine under the same current FLOP meter. The final mean-source addition also improves the official score by **2.05%** relative to our covariance-only entry. These are different comparisons. The much larger 28.50% improvement over our earlier incumbent is predominantly associated with adopting the public cost engine; we do not claim that improvement as our independent contribution. We document training, resource measurements, unsuccessful alternatives, and the limits of the evidence. Private final evaluation has not yet occurred.

1 Problem and score

For a fixed, bias-free, width-1024, depth-16 ReLU MLP with known weights,

$$h_0 = X \sim \mathcal{N}(0, I), \qquad h_\ell = \mathrm{ReLU}(W_\ell h_{\ell-1}), \qquad m_\ell = \mathbb{E}_X[h_\ell], \tag{1}$$

the estimator returns all layer means; the scored error is on layer 16. For network j, with counted compute C_j and budget $B = 2^{41}$,

$$s_j = \frac{\|\widehat{m}_{16,j} - m_{16,j}\|_2^2}{1024} \max\{0.1, C_j/B\}, \qquad S = \frac{1}{N}\sum_j s_j. \tag{2}$$

The score is the *mean of per-network products*; multiplying mean error by mean compute is generally incorrect. The challenge uses a separate fresh private evaluation for final rankings [4, 5].

2 Published foundation and our changes

Wu et al. [1] develop cumulant and Hermite methods for propagating activation distributions through random MLPs. Our submission belongs to that family. It does not run Monte Carlo input samples at inference.

The direct implementation base is LordProcciuto's MIT-licensed `e30_a`, commit `3d038e9`, derived from 504aldo's MIT V29 engine [3, 2]. We retain its approximate mean/covariance propagation, factored third-order history, regenerated fourth-order information, pruning, fixed transported bases, matrix-product engineering, and existing fitted 87-feature mean correction. Those are credited prior work. The final archive preserves both authors' MIT notices.

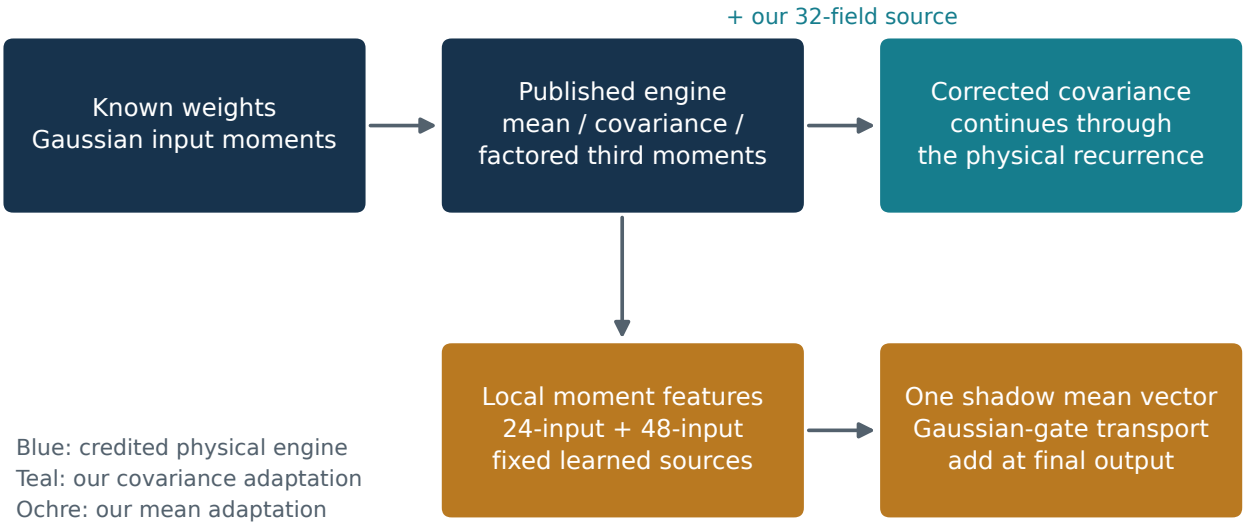


Figure 1: The two additions act at different places. Our covariance source changes the continuing physical state. The new mean-source vector follows that trajectory but does not feed back into it.

Component	Contribution claimed in this report
Physical covariance	A 32-field defect dictionary, 96 fitted coefficients across three layer groups, including marginal, mixed, and quadratic third-cumulant shapes; an algebraic regrouping of its evaluation.
Mean innovation	A trace-informed local feature representation, training on layerwise error innovations, and one transported response vector.
Final combination	Equal averaging of independently trained 24-input and 48-input source models, with their normalization and clipping preserved in a single batched call.

These are adaptations of established cumulant, response, and learned-closure ideas. We make no claim to have invented cumulant propagation, Hermite expansions, residual learning, or model averaging. The covariance dictionary is mechanistically motivated; its coefficients and the mean-source functions are learned. The submission is therefore not a parameter-free deduction from the published equations.

3 A covariance-defect source inside the recurrence

At a preactivation, write mean μ_i, variance $v_i = \sigma_i^2$, off-diagonal covariance c_{ij}, and $\alpha_i = \mu_i/\sigma_i$. Define the Gaussian-smoothed ReLU derivatives

$$f_i^{(0)} = \mu_i\Phi(\alpha_i) + \sigma_i\phi(\alpha_i), \quad f_i^{(1)} = \Phi(\alpha_i), \quad f_i^{(r)} = \frac{(-1)^{r-2}\,\mathrm{He}_{r-2}(\alpha_i)\phi(\alpha_i)}{\sigma_i^{r-1}} \quad (r \geq 2). \tag{3}$$

Here ϕ, Φ are the standard Gaussian density and CDF, and He_r are probabilists' Hermite polynomials. These are derivatives of a Gaussian convolution, so no ordinary higher derivative of the nonsmooth ReLU needs to be evaluated.

For jointly Gaussian coordinates, the off-diagonal covariance expansion is

$$\mathrm{Cov}_G(f(Z_i), f(Z_j)) = \sum_{r\geq 1} \frac{c_{ij}^r}{r!} f_i^{(r)} f_j^{(r)}. \tag{4}$$

A formal first-order marginal third-cumulant perturbation suggests fields

$$S_{r,ij}^{(3)} = \frac{c_{ij}^r}{6r!}\left[k_{3,i} f_i^{(r+3)} f_j^{(r)} + k_{3,j} f_j^{(r+3)} f_i^{(r)}\right]. \tag{5}$$

Mixed third cumulants instead supply shapes beginning with $\frac{1}{2}T_{iij} f_i^{(2)} f_j^{(1)} + (i \leftrightarrow j)$. Products of third cumulants supply additional quadratic shapes. The dictionary also includes diagonal variance sources and fourth-cumulant terms; Appendix A specifies all 32 fields.

These expressions motivate a *defect dictionary*, not a list of proven missing terms. The base already includes some of their effects. Adding each textbook coefficient with unit strength would double-count contributions. Instead, the actual postactivation covariance is changed by

$$C_\ell^+ \leftarrow C_{\ell,\mathrm{base}}^+ + \sum_{a=1}^{32} \beta_{g(\ell),a} S_a, \qquad g(\ell) = \min\{\lfloor(\ell-1)/5\rfloor, 2\}. \tag{6}$$

The correction is used after layers 1–15; no covariance is needed after the final mean readout. Its diagonal is also added to the stored variance vector. This state subsequently affects transport and nonlinear moment production.

Fitting. The 96 coefficients were fitted previously on 32 exposed training networks using a Gaussian mean/covariance response model. The loss combines normalized local mean innovations and terminal mean error, with diagonal ridge regularization $\lambda = 10^{-3}$. The target trajectory then included an earlier mean corrector. All 96 coefficients were transferred unchanged to the newer public engine; they were not refitted for submission 336809. The response is an approximation to the derivative of the full algorithm, so actual integrated evaluations, rather than the fit alone, establish the reported gains.

Evaluation cost. Terms sharing a power of c_{ij} or a cumulant slice are grouped. Sums of separable kernels become small products AB^T, reducing Python operations and repeated elementwise work. This is an algebraic regrouping; floating-point evaluation order can change.

4 Learning and transporting local mean innovations

Let b_ℓ be the base prediction along the covariance-corrected trajectory, and let $e_\ell = m_\ell - b_\ell$ be its training error. Set $J_\ell = \mathrm{diag}(\Phi(\alpha_\ell))W_\ell$. Our supervised target is

$$u_\ell = e_\ell - J_\ell e_{\ell-1}, \qquad \ell = 2, \dots, 16. \tag{7}$$

This separates locally generated error from error already carried through the linearized mean map. For any chosen J_ℓ, the identity is exact if the true errors are supplied; interpreting J_ℓ as the response of the full cumulant estimator is an approximation.

At inference, a source model predicts $\widehat{u}_\ell$ from current moments, and one shadow vector evolves as

$$r_1 = 0, \qquad r_\ell = J_\ell r_{\ell-1} + \widehat{u}_\ell. \tag{8}$$

The new vector does not change the physical mean, covariance, or cumulant history. It is added at the last layer. In the exact code order the final output is $[b_{16}^{\mathrm{core}} + r_{16}]_+ + d_{16}$, where d_{16} is the public engine's separately restored contribution from pruned neurons. Earlier returned layer means remain the physical predictions.

Features. The local source receives 24 features: normalized mean, skewness and fourth cumulant; relative log variance; eight pairwise correlation/cumulant summaries; three layer/global scalars; and nine signed mean/trace projections. Appendix B gives their formulas. The context source adds the within-layer neuron average of each local feature, giving 48 inputs. At the final layer, the pair and trace features are set to zero because the public engine omits those expensive final-layer slices.

For either input vector x_i, a model is

$$\begin{aligned} z_i &= \mathrm{clip}((x_i - a)/s, -8, 8), \\ \widehat{u}_{\ell,i} &= 10^{-3}\sigma_i\phi(\alpha_i)\left[w_{h(\ell)}^T \tanh(Hz_i + b) + d_{h(\ell)}\right]. \end{aligned} \tag{9}$$

Each has 32 hidden units and four output heads, selected for layers 2–6, 7–11, 12–15, and 16. The prefactor provides activation units and suppresses corrections far from the ReLU boundary. It does not prove that the learned source equals an omitted cumulant term.

The shipped combination. The 24-input model was trained on 32 networks; the 48-input model was trained on 64, including those 32. The submission uses their fixed equal average. No mixture weight was fitted. Because the models have different means and scales, their overlapping features are duplicated before normalization: 72 coordinates feed a block-diagonal 64-unit hidden layer, with halved output weights. There are 2,632 independently learned neural parameters before this deterministic combination, plus feature statistics. Batching preserves the mathematical function, not bitwise equality to separate float32 evaluation. The recorded four-head feature test found maximum absolute difference 7.15×10^{-7} before the amplitude prefactor. Exact-byte replay was checked separately for the shipped package.

5 Measured performance and attribution

The three recent entries were graded successfully on 17 October and their scores were rechecked on 18 October. Table 1 reports official public results. Only **336809** is the submission associated with this writeup; the other IDs are comparison evidence.

Submission	Method	Raw MSE	Adjusted score
335391	Earlier engine + covariance	2.01272×10^{-8}	4.15562×10^{-9}
336778	Public e30 + our covariance	1.98938×10^{-8}	3.03329×10^{-9}
336785	Above + trace source	1.95309×10^{-8}	2.98282×10^{-9}
336809	**Above + equal source average**	1.94468×10^{-8}	2.97111×10^{-9}

Table 1: Official public grades. The older entry was evaluated before the latest meter update; this historical sequence is not a controlled cost ablation.

The final score is 28.50% below our earlier incumbent. Raw error falls only 3.38%; the ratio of aggregate score/error ratios is 0.73998. That second quantity is an error-weighted multiplier comparison, *not* the mean FLOP ratio. The main cost-engine improvements belong to the public authors.

To isolate the incremental effect of our adaptation, Table 2 uses the same four exposed development networks and the same current runtime for every row. There is no official unmodified-base submission in this controlled comparison.

Method	Raw MSE 10^{-8}	FLOPs 10^{9}	Adjusted 10^{-9}	Reduction vs public
Public e30 engine	2.11094	336.173	3.22335	—
+ our covariance	2.02488	338.069	3.10976	3.52%
+ trace source	2.00623	338.606	3.08614	4.26%
+ equal source average	1.99919	338.738	3.07676	4.55%

Table 2: Matched development ablation on rows 750, 976, 772, 795; FlopScope 0.12.3 / WhestBench 0.16.2. These are adaptively reused development inputs, not a fresh held-out test. FLOPs are unweighted means; adjusted scores are computed per network before averaging.

The physical covariance addition costs about 0.56% extra arithmetic in this comparison. The complete adaptation costs about 0.76% more than the public engine and reduces raw error by 5.29%. Its adjusted benefit therefore survives the measured compute charge.

The newer mean corrections also have a wider comparison: on 12 development networks the trace source improves adjusted score by 2.23% over covariance alone, winning 11/12; officially the improvement is 1.66%. The final equal average improves another 0.170% locally and 0.393% officially. Combined, the official mean-source gain over covariance alone is 2.05%. This supports a modest additional benefit, not a claim that the learned sources explain the large historical score reduction.

6 Training and strength of the evidence

Network-level splits. The covariance and original trace source use 32 exposed public training networks. The context source uses those 32 plus 32 additional public training networks. Four folds hold out whole networks, not individual neurons. Final assets use all networks in their stated training sets. The 12 submission-facing development rows are disjoint from those fitting rows, but were repeatedly used during the research campaign. Our protected final local blind block was not used in this task or to choose these submitted assets. Local errors use the organizer's reference means, not independently known exact expectations; finite-reference error was not subtracted.

Mean-source training. Both models use 1,000 Adam updates with learning rate 0.01, batches of 32,768 neuron/layer examples, gradient norm clipping at 5, and parameter-mean-square penalty 10^{-4}. The source-error loss is normalized by the training RMS innovation at each layer, floored at 10^{-8}. Feature standard deviations are floored at 10^{-12}. Normalization and layer loss scales are computed from the training fold. The output layer is initialized to zero; final-fit initialization and batch generator seeds are 172018 and 172118. Training uses PyTorch offline; inference uses fixed JSON assets and permitted FlopScope operations.

Source	Response-model evaluation	Raw error ratio	Wins
Trace24	Four folds, original 32 networks	0.97486	26/32
Context48	Four folds, expanded 64 networks	0.96752	56/64
Frozen Trace24	Additional 32 trajectories	0.96352	30/32

Table 3: Ratios to the covariance-only trajectory in the offline response calculation. They are conditional/adaptive evidence, not integrated native cross-validation or independent private-suite estimates.

The covariance coefficients already depend on the original 32 networks, so holding out a network for the source model does not make the entire pipeline independent of that network. Architecture choices also used development feedback. Offline response results can differ from a full estimator run, especially when changing the physical state or its pruning decisions.

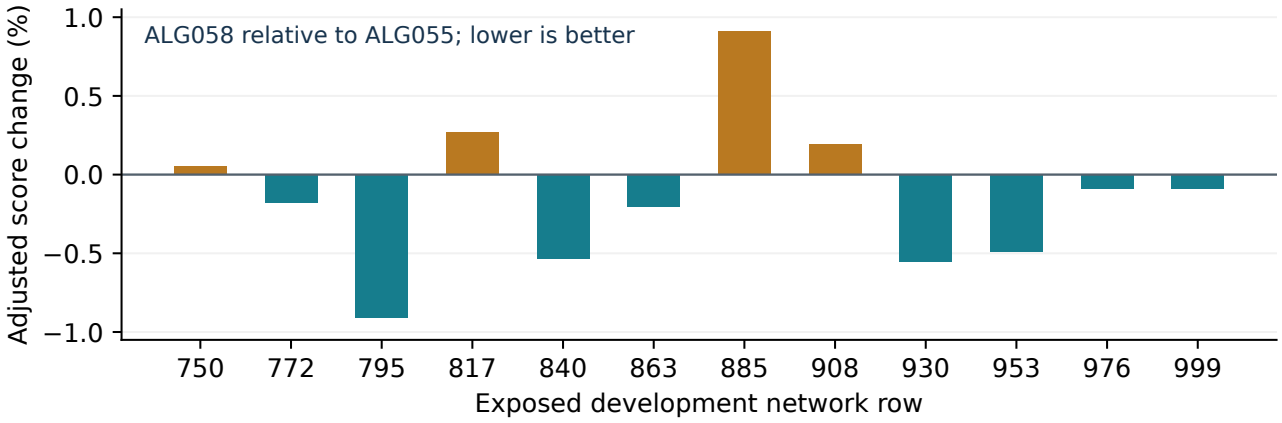


Figure 2: The final combination wins 8/12 exposed development comparisons against the simpler trace model. The plotted percentage is per network; the pooled adjusted-score reduction is 0.170%.

The archived bootstrap support for a lower pooled mean in this last comparison is 0.893; its 90% ratio interval is [0.99624, 1.00058]. Because these inputs were adaptively reused, this is descriptive resampling support, not an 89.3% probability of winning the final private evaluation. The official public improvement provides additional feedback, but that leaderboard also guided development. We do not infer a confidence interval for the private rank.

7 Failures, limitations, and resource behavior

Corrections are trajectory-dependent. Transferring our older fitted mean corrections directly to the cheaper public engine gave four-network adjusted scores around 3.40–3.53×10^{-9}, worse than the public engine's 3.22×10^{-9}. Refitting the local mean source on the new covariance-corrected trajectory worked. Conversely, refitting all covariance coefficients on the new trajectory did not improve the tested conditional cross-validation. There is no general guarantee that a more recently fitted or more flexible corrector is better.

A promising small pilot can reverse. A contextual source split between physical mean feedback and the shadow vector improved a four-network pilot by 1.08% but lost by 0.53% on the 12-network comparison. It was not submitted. The 64-network context source alone was effectively tied with the trace source on the larger comparison; the retained gain comes from their combination. We did not establish why that averaging helps. Error complementarity is a plausible interpretation, not an isolated mechanism.

More moment information can cost too much. Exploratory prototypes using ARC's carried-fourth equations and a public scale-extraction proposal [7] showed roughly 15–20% raw-error reductions on two training networks. A literal rank-32 compression/transport implementation projected about 51% extra arithmetic, whereas its measured raw reduction would tolerate only about 18% extra compute above the score floor. This combines research accuracy with separately metered kernels; it is neither an integrated score nor a lower bound on the method family. These prototypes are not part of submission 336809. They remain open research routes requiring substantial cost reductions.

Operational improvement mattered. Evaluating the two source models separately exceeded the 0.4-second residual-time cap on one local archive case (0.401164 seconds). That version was not uploaded. Batching the unchanged models produced the accepted version and passed the recorded full-shape checks. Table 4 reports observed maxima, not guarantees for every future network or host.

Resource	Observed for ALG058-V001	Limit
Counted FLOPs, per network	328.222–351.376×10^{9}	2^{41}
Residual wall time, maximum	0.387661 s	0.4 s
Predict wall time, maximum	44.285 s	120 s
Setup time, maximum	0.047888 s	5 s
Process peak RSS	6,086,852,608 bytes	2^{33} bytes

Table 4: Preserved pilot and clean-archive qualification evidence under the current local runtime. Twelve distinct full-size inputs plus repeated-input checks; all recorded checks pass.

The package returns finite 16×1024 predictions and has a separate shape-generic smoke-test path inherited from the base. It does not establish that the fitted corrections generalize to other widths, depths, activations, or weight distributions. Resource failures and accuracy losses were recorded separately throughout development.

8 Reproducibility, provenance, and disclosure

Exact submitted artifact. The report concerns submission **336809**, ALG058-V001. Its accepted archive is identified by SHA-256:

```
2c28711f6d44709182886eb6e1535503af415
ecd97721b46868b39f2ff0660db
```

The two lines concatenate to one digest. The estimator and neural-asset digests, source map, and training row lists are in Appendix C. The archive already held by AIcrowd contains the executable estimator, fixed asset, metadata, README, and preserved licenses. A later documentary clarification must not be confused with changing that accepted code.

Local qualification used Python 3.12.3, WhestBench 0.16.2, FlopScope and FlopScope-server 0.12.3, and NumPy 2.2.6. The evaluator was updated to these meter versions before the three recent submissions. The algorithm's numeric inference operations use FlopScope; no runtime training, network access, hidden grader information, or target lookup is used. Exposed challenge training labels and weights remain in the research data store rather than in the submitted estimator.

Development. Earlier work investigated sampling/variance reduction, Gaussian and higher-cumulant closures, covariance response models, and learned corrections. The retained implementation combines a published efficient engine with the covariance fields developed in our earlier recurrence and the two newer source models. It reflects both mathematical derivation and empirical fitting. Most explored alternatives were not incorporated. The report focuses on the accepted algorithm and the evidence needed to assess its added value rather than presenting every research attempt as a finding.

LLM assistance and author responsibility. LLM coding agents performed most of the code writing, mathematical exploration, experiment orchestration, literature inspection, and report drafting. Substantial earlier work used GPT-5.6 Sol; later work primarily used GPT-6 Astra at extra-high effort, running locally through the ChatGPT Desktop application. A recorded Ultra-effort trial used parallel sub-agents; smaller contributions came from Fable and Opus sessions. Tristan Miano set objectives, directed the research strategy, authorized tasks and submission attempts, and reviewed progress. This draft is prepared for his review. Claims of identities are supported by the displayed algebra or recorded numerical checks; explanations of empirical success that were not isolated experimentally are labeled as hypotheses. The participant is responsible for the final submitted report.

Scope of originality. Public cumulant machinery, its fitted reference tables, pruning, and core cost engineering are not ours. Our claim is the specific covariance-defect adaptation, local-source construction, and their measured incremental value in this estimator. We do not claim academic priority for the general methods. The organizer's public-code clarification requires substantial own changes; this report provides evidence for review and does not claim advance eligibility approval [6]. The official result is an improvement on our prior entries, not proof that this is the best published estimator or a future prize winner.

References

[1] W. Wu, V. Lecomte, M. Winer, G. Robinson, J. Hilton, and P. Christiano. *Estimating the expected output of wide random MLPs more efficiently than sampling*. arXiv:2605.05179v2, 2026. https://arxiv.org/abs/2605.05179v2.

[2] 504aldo. *whest-p2-cumulant-k3*, V29. MIT-licensed public implementation. https://github.com/504aldo/whest-p2-cumulant-k3.

[3] LordProcciuto. *whestbench-lordprocciuto*, e30_a and accompanying writeup. MIT; commit 3d038e9f48a97671878ee7bbef02e193f128b61c. https://github.com/siming767-creator/whestbench-lordprocciuto.

[4] ARC / AIcrowd. *Official challenge rules*, accessed 18 October 2026. https://www.aicrowd.com/challenges/arc-white-box-estimation-challenge-2026/challenge_rules.

[5] AIcrowd. *Phase 2 update*, 17 October 2026. https://discourse.aicrowd.com/t/18251.

[6] AIcrowd. *Eligibility of solutions built on another participant's MIT-licensed forum code*, 5 October 2026. https://discourse.aicrowd.com/t/18240.

[7] il.miscusi. *Removing the global scale mode makes a carried fourth-cumulant core pay*, Phase 2 writeup for submission 336317, 15 October 2026. https://discourse.aicrowd.com/t/18248.

A The covariance dictionary

This appendix specifies the unfused 32-field construction. The submitted code evaluates its weighted sum in grouped kernels. For distinct i, j, abbreviate $c = c_{ij}$, $a = k_{3,i}$, $b = k_{3,j}$, $t = T_{iij}$, $u = T_{jji}$, $q_i = k_{4,i}$, and $\gamma_i = q_i/v_i^2$. Write $F_{pq} = f_i^{(p)} f_j^{(q)}$, with derivatives from Eq. (3). $K_{31,ij}$ and $K_{22,ij}$ are the base's available mixed fourth-cumulant approximations, not observed true cumulants. In the code these are `wk431` and `wk4m`, respectively; k_3, T, k_4 use `D3`, `D21`, and `g4row`. Each listed field has an independent coefficient in each of three layer bins.

Six basic fields. Let $V_i^G = (\mu_i^2 + v_i)\Phi(\alpha_i) + \mu_i\sigma_i\phi(\alpha_i) - (f_i^{(0)})^2$. The first two fields are diagonal matrices with entries V_i^G and $V_i^G\gamma_i$. The four off-diagonal fields are

$$\tfrac{1}{2}cF_{11}(\gamma_i + \gamma_j), \qquad \frac{c^r}{r!}F_{rr} \quad (r = 2, 3, 4). \tag{10}$$

All their unspecified diagonal entries are zero.

Eight marginal fields. Two diagonal fields are

$$\frac{k_{3,i}}{3}[f_i^{(2)} - f_i^{(0)} f_i^{(3)}], \qquad \frac{k_{4,i}}{12}[f_i^{(3)} - f_i^{(0)} f_i^{(4)}]. \tag{11}$$

For $r = 1, 2, 3$, six off-diagonal fields are

$$\frac{c^r}{6r!}(aF_{r+3,r} + bF_{r,r+3}), \qquad \frac{c^r}{24r!}(q_iF_{r+4,r} + q_jF_{r,r+4}). \tag{12}$$

Eight mixed/reference fields. The first seven are

$$\begin{aligned} &\frac{c^r}{2r!}(tF_{r+2,r+1} + uF_{r+1,r+2}), && r = 0, 1, 2, \\ &\frac{K_{22,ij}c^r}{4}F_{r+2,r+2}, && r = 0, 1, \\ &\frac{c^r}{6}(K_{31,ij}F_{r+3,r+1} + K_{31,ji}F_{r+1,r+3}), && r = 0, 1. \end{aligned} \tag{13}$$

The eighth is $C^+_{\text{base}} - G_8$ off the diagonal, where $G_{8,ij} = \sum_{r=1}^{8} c^r F_{rr}/r!$. This allows a fitted correction to the difference between the base pair closure and its order-eight Gaussian reference. The unfused helper clips normalized correlation near ± 1; the shipped fused kernel evaluates the corresponding polynomial directly from the carried covariance. Exact shipped code is the authority at numerical boundaries.

Ten quadratic third-cumulant fields. The first two are

$$\frac{c}{72}(a^2F_{71} + b^2F_{17}), \qquad \frac{cab}{36}F_{44}. \tag{14}$$

For each $r = 0, 1$, four more are

$$\begin{aligned} &\frac{c^r}{8}(t^2F_{r+4,r+2} + u^2F_{r+2,r+4}), && \frac{c^r tu}{4}F_{r+3,r+3}, \\ &\frac{c^r}{12}(atF_{r+5,r+1} + buF_{r+1,r+5}), && \frac{c^r}{12}(btF_{r+2,r+4} + auF_{r+4,r+2}). \end{aligned} \tag{15}$$

Their diagonals are zero. The formulas motivate signed residual shapes; they do not assert that each missing term can be uniquely identified from fitted coefficients. The 96 shipped values exactly match the preserved all-32-network fit. They are included in the machine-readable evidence rather than rounded into this report.

B Mean-source features and training objective

Define $R_{ij} = c_{ij}/(\sigma_i\sigma_j)$, $P_{ij} = T_{iij}/(v_i\sigma_j)$, $g_{3,i} = k_{3,i}/\sigma_i^3$, $g_{4,i} = k_{4,i}/v_i^2$, and $\tau_i = \sum_j T_{jji} + k_{3,i}$. Angle brackets denote a neuron average at the current layer. Superscript $\circ r$ means entrywise power.

Group	Features (in the implemented order)
Local (4)	$\alpha_i,\ g_{3,i},\ g_{4,i},\ \log(v_i/\langle v\rangle)$.
Pair (8)	$\sum_j R_{ij}^2,\ \sum_j R_{ij}^3,\ \sum_j R_{ij}^4,\ \sum_j P_{ij}^2,$ $\sum_j P_{ji}^2,\ \sum_j R_{ij}P_{ij},\ \sum_j R_{ij}P_{ji},\ (R^{\circ 2}g_4)_i.$
Global (3)	$\langle g_4\rangle,\ \langle g_3^2\rangle,\ (\ell-1)/15.$
Trace/signed (9)	$\sum_j P_{ij},\ \sum_j P_{ji},\ (R\alpha)_i,\ (P\alpha)_i,\ (P^T\alpha)_i,$ $((R\circ P)\alpha)_i,\ ((R\circ P^T)\alpha)_i,\ \tau_i/(\sigma_i\langle v\rangle),$ $\langle\tau\mu\rangle/(\langle\mu^2\rangle\langle v\rangle + 10^{-30}).$

The 48-input model appends 24 neuron-averaged features. Its statistics are computed from the features as represented by the public engine, including pruning conventions. The final layer uses zero pair and signed/trace blocks. The 24-input model and 48-input model have different normalization statistics; duplicating local coordinates in the batch preserves both sets of clipping boundaries. Replacing them with a common normalizer would be a different model.

For source training, with training-fold layer RMS s_ℓ, the minibatch objective is

$$\mathcal{L}(\theta) = \mathrm{mean}_{(j,\ell,i)\in\mathcal{B}}\left[\frac{\widehat{u}_{j\ell i}(\theta) - u_{j\ell i}}{s_\ell}\right]^2 + 10^{-4}\frac{\|\theta\|_2^2}{\#\theta}. \tag{16}$$

The 24-input final fit has 932 learned parameters; the 48-input model has 1,700. The merged 64-unit representation contains structural zeros and a combined output bias, so counting every stored array element as an independent learned parameter would overstate the fitted degrees of freedom.

The innovation identity check in the research code starts from the observed first-layer error and reconstructs the final observed error to numerical precision. Deployment starts its additional response at zero and predicts the later innovations. Thus that identity check validates the construction of training targets; it does not prove the deployed response is exact or that the first-layer error is identically zero.

For covariance fitting, let $R_{j\ell}\beta$ denote the approximate mean response and $B_{j\ell}\beta = R_{j\ell}\beta - J_{j\ell}R_{j,\ell-1}\beta$. The linear fitting system combines

$$\mathrm{mean}_{j,\ell,i}\,\frac{(B_{j\ell}\beta - u_{j\ell})_i^2}{s_\ell^2} + \mathrm{mean}_{j,i}\,\frac{(R_{j,16}\beta - e_{j,16})_i^2}{s_{16,\mathrm{end}}^2} + 10^{-3}\beta^T\,\mathrm{diag}(G)\beta, \tag{17}$$

where G is the normalized design Gram matrix, with a 10^{-30} floor on its diagonal in the implementation. The coefficients were selected from the local-plus-terminal objective; terminal-only fitting was a diagnostic alternative. The later full-state evaluation is not identical to this linear response model.

C Artifact identity and evidence map

The accepted estimator and fixed neural asset have the following SHA-256 digests (join the wrapped lines):

```
estimator.py:
6c295943161d1365afa042948770e451091d
6fe16706ee7728505ece62b4ab44

nn_asset.json:
bd824f7c1a7e4d1392e0b8507931e973237e
b3ca02cabc9a07b66970d2c8e856
```

The accepted archive, source files, current working copies, and stored qualification hashes were compared on 18 October 2026; all matched. This was an identity audit using existing evidence, not another numerical qualification campaign. The live API still reported submission 336809 as successfully graded with the scores shown in Table 1.

Claim / component	Repository evidence or code location
Accepted implementation	`submissions/phase2_alg058_v001/estimator.py`; mean-source call around lines 3287–3311; covariance update around 3540; fused helper `_fused_cov_delta`.
Covariance fit	`experiments/theory/t170/fit_quadratic.py` and `experiments/theory/t170/P27_FIT/RESULT.json`.
Source models	`experiments/theory/t172/train_source.py`, `train_expanded_source.py`, and `build_anchor_merged.py` (last two in the same directory).
Matched public-base comparison	`experiments/theory/t172/runs/`, probes P02_e30, P09_fused_cov, P19_new_trace, P82_anchor_merged.
Official and qualification records	`experiments/theory/t172/candidates/`, ALG054-V000, ALG055-V000, ALG058-V001.
Report evidence index	`experiments/finalization/t173/WRITEUP_EVIDENCE.json`, `PACKAGE_IDENTITY_AUDIT.json`, and `CURRENT_GRADES.json` (all in that directory).

These repository paths identify retained evidence; they are not claims that the entire working repository or its research data has already been publicly released. The AIcrowd submission ID resolves the accepted executable package. Any public evidence release should include the relevant scripts, manifests, predictions/summaries, and licenses, without republishing private challenge weights or labels.

Training rows. Labels come from the organizer's `aicrowd/arc-whestbench-public-2026` Full dataset; the original layer-label manifest pins revision `ce926b7d1ced65faa668c01b94a43f8c4f43783d`. The original 32 exposed dataset positions were

> 0, 16, 32, 48, 64, 80, 97, 113, 129, 145, 161, 177, 193, 209, 225, 241, 258, 274, 290, 306, 322, 338, 354, 370, 386, 402, 419, 435, 451, 467, 483, 499.

The additional 32 were

> 8, 23, 39, 55, 70, 86, 102, 117, 133, 148, 164, 180, 195, 211, 227, 242, 259, 273, 289, 305, 320, 336, 352, 367, 383, 398, 414, 430, 445, 461, 477, 492.

The submission-facing development order was 750, 976, 772, 795, 817, 840, 863, 885, 908, 930, 953, 999. These are dataset row identifiers, not seeds or network identifiers used by the submitted inference algorithm.